

ECO101 - Introduction to Microeconomics

Department of Economics and Social Sciences (ESS), Brac University

Perfect Competition

Lecture: 7

Reading

- Economics by Michael Parkin, 10th edition
 - Chapter: 12

Outline

- Define perfect competition
- Explain how a firm makes its output decision and why it sometimes shuts down temporarily and lays off its workers
- Explain how price and output are determined in a perfectly competitive market
- Explain why firms enter and leave a competitive market and the consequences of entry and exit

Introduction - Markets and Competitive Environment

Economists identify four market types:

1. Perfect competition
2. Monopolistic competition
3. Oligopoly
4. Monopoly

Perfect Competition

Perfect competition is a market in which

- Many firms sell identical products to many buyers
- There are no restrictions on entry into the market/industry
- Established firms have no advantage over new ones
- Sellers and buyers are well informed about prices

Farming, fishing, wood pulping and paper milling, the manufacture of paper cups and shopping bags, grocery and fresh flower retailing, photo finishing, lawn services, plumbing, painting, dry cleaning, and laundry services are all examples of highly competitive industries

Price Takers

- Firms in perfect competition are price takers
- A price taker is a firm that cannot influence the market price because its production is an insignificant part of the total market
- In perfect competition, each firm produces a (homogenous) good that has no unique characteristics, so if a particular firm charges higher price than the market price then consumers will not buy from that firm
- Consider what would happen, if the firm decides to sell at a cheaper price than the market price?

Economic Profit and Revenue

- A firm's goal is to maximize economic profit, which is equal to total revenue minus total cost

$$\text{Profit} = \text{TR} - \text{TC}$$

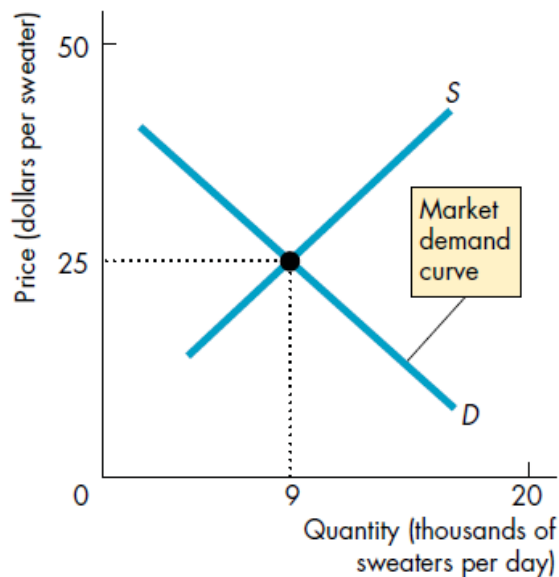
- A firm's total revenue equals the price of its output multiplied by the number of units of output sold (price $\times$ quantity)
- Marginal revenue is the change in total revenue that results from a one-unit increase in the quantity sold
- Marginal revenue is calculated by dividing the change in total revenue by the change in the quantity sold
- Average revenue is calculated by dividing total revenue by the quantity sold

Economic Profit and Revenue

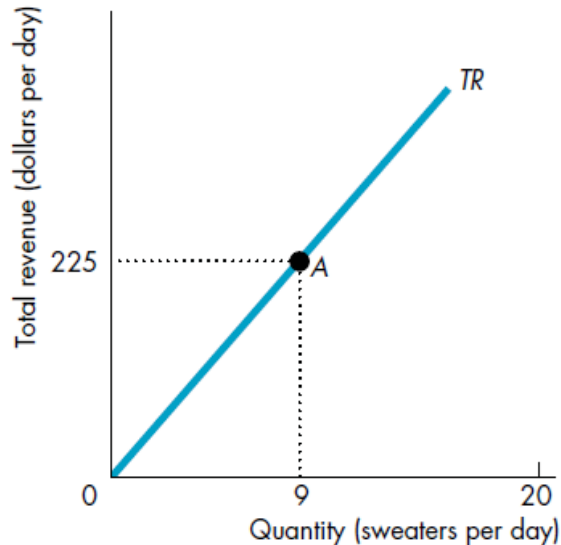
Quantity sold (Q) (sweaters per day)	Price (P) (dollars per sweater)	Total revenue ($TR = P \times Q$) (dollars)	Marginal revenue ($MR = \Delta TR / \Delta Q$) (dollars per additional sweater)
8	25	200	 25
9	25	225	 25
10	25	250	

Demand, Price, and Revenue in Perfect Competition

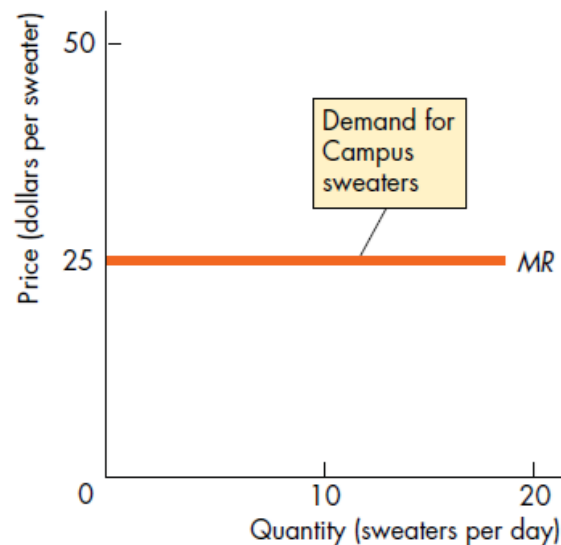
FIGURE 12.1 Demand, Price, and Revenue in Perfect Competition



Sweater Market



Sweater Firm's Total Revenue



Sweater Firm's Marginal Revenue

Demand, Price, and Revenue in Perfect Competition

- Market demand and market supply determine the market price (and quantity)
- The firm can sell any quantity it chooses at the market price
- So, the demand curve for the firm's product is a horizontal line at the market price, the same as the firm's marginal revenue curve
- Hence, $MR = P$
- A horizontal demand curve illustrates a perfectly elastic demand, so the demand for the firm's product is perfectly elastic

The Firm's Output Decision

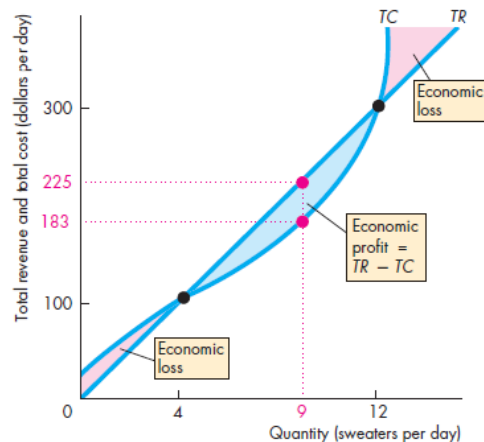
- The goal of the competitive firm is to maximize economic profit, given the constraints it faces
- From the firm's cost curves and revenue curves, we can find the output that maximizes the firm's economic profit
- Economic profit equals total revenue minus total cost

Total Revenue, Total Cost, and Economic Profit

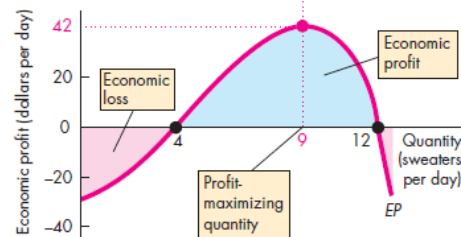
Quantity (Q) (sweaters per day)	Total revenue (TR) (dollars)	Total cost (TC) (dollars)	Economic profit (TR - TC) (dollars)
0	0	22	-22
1	25	45	-20
2	50	66	-16
3	75	85	-10
4	100	100	0
5	125	114	11
6	150	126	24
7	175	141	34
8	200	160	40
9	225	183	42
10	250	210	40
11	275	245	30
12	300	300	0
13	325	360	-35

- Firm maximizes its economic profit by producing 9 sweaters a day
- At outputs of less than 4 sweaters and more than 12 sweaters a day, the firm would incur an economic loss
- At either 4 or 12 sweaters a day, the firm would make zero economic profit (break-even point)

Total Revenue, Total Cost, and Economic Profit



(a) Revenue and cost



(b) Economic profit and loss

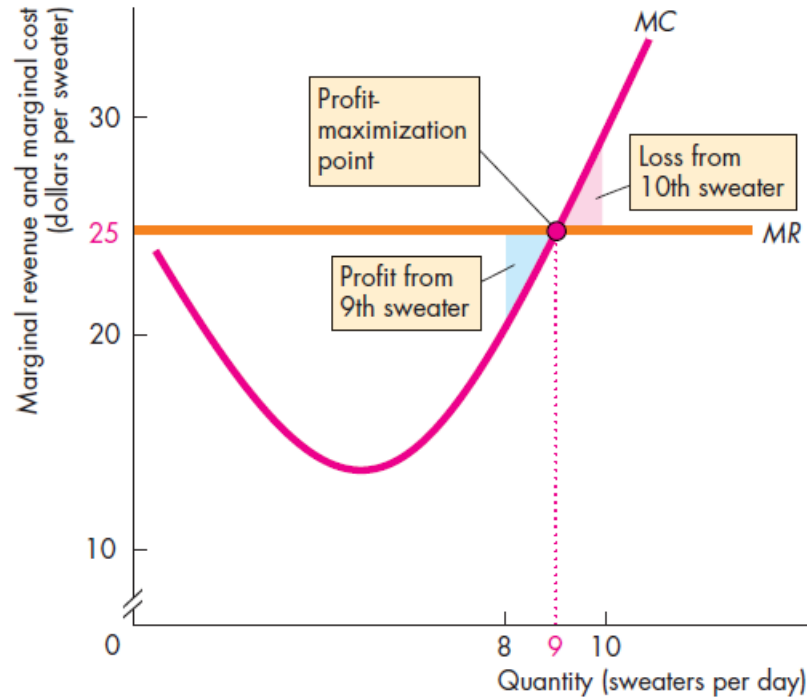
Maximizing Profits: Marginal Analysis

- Another way to find the profit-maximizing output is by comparing marginal revenue, MR, with marginal cost, MC
- As output increases, the firm's marginal revenue is constant but its marginal cost eventually increases
- If marginal revenue exceeds marginal cost, an increase in output increases economic profit
- If marginal revenue is less than marginal cost, an increase in output decreases economic profit
- If marginal revenue equals marginal cost, economic profit is maximized

Maximizing Profits: Marginal Analysis

Quantity (Q) (sweaters per day)	Total revenue (TR) (dollars)	Marginal revenue (MR) (dollars per additional sweater)	Total cost (TC) (dollars)	Marginal cost (MC) (dollars per additional sweater)	Economic profit (TR - TC) (dollars)
7	175	25	141	19	34
8	200	25	160	23	40
9	225	25	183	27	42
10	250	25	210	35	40
11	275		245		30

Maximizing Profits: Marginal Analysis



→ Profit maximizing level of output is 9 sweaters per day

→ $P = MR = MC$

Temporary Shutdown Decision

- A firm maximizes profit by producing the quantity at which marginal revenue (price) equals marginal cost
- But suppose that at this quantity, price is less than average total cost and firm incurs an economic loss
- If the firm expects the loss to be permanent, it goes out of business
- But if it expects the loss to be temporary, the firm must decide whether to shut down temporarily and produce no output, or to keep producing
- To make this decision, the firm compares the loss from shutting down with the loss from producing and takes the action that minimizes its loss

Temporary Shutdown Decision

Loss Comparison

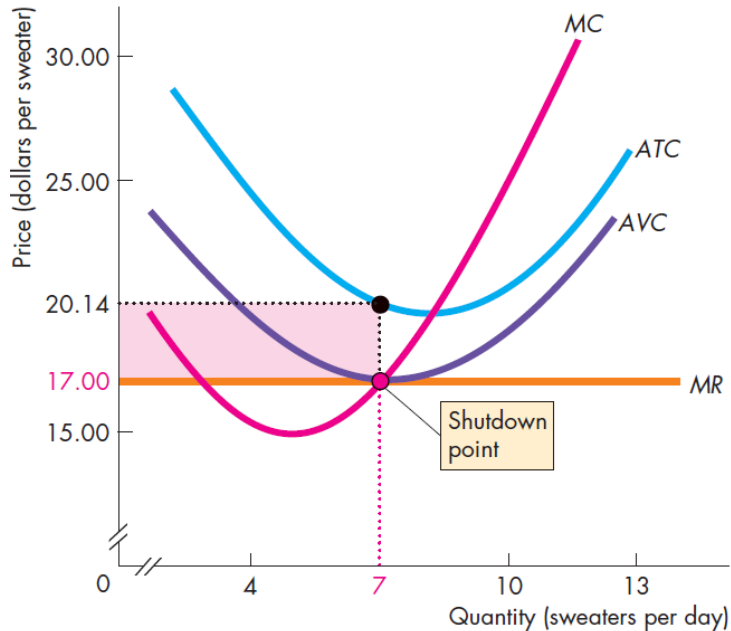
Economic Loss = $TFC + TVC - TR$

$$= TFC + (AVC - P) \times Q$$

- If the firm shuts down, it produces no output ($Q = 0$) and only pay its fixed costs, so its economic loss equals total fixed cost
- If the firm produces, then in addition to its fixed costs, it incurs variable costs, but it also receives revenue
- Its economic loss equals total fixed cost plus total variable cost minus total revenue
- If $TVC > TR$, this loss exceeds TFC and the firm shuts down
- Equivalently, if $AVC > P$, this loss exceeds TFC and the firm shuts down

The Shutdown Point

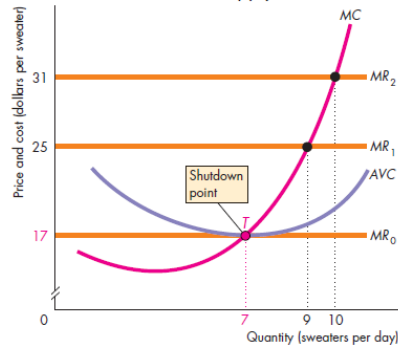
FIGURE 12.4 The Shutdown Decision



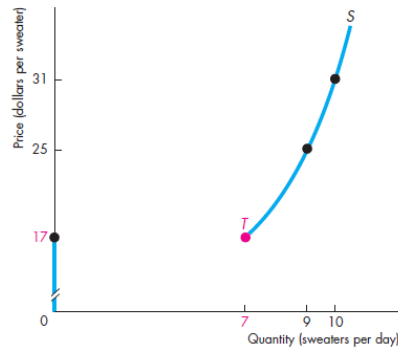
- Firms shut down temporarily, if price falls below the minimum of the average variable cost (AVC)
- At shutdown point firm is minimizing its loss and incurs a loss equal to TFC
- At prices above minimum AVC but below ATC, the firm produces the loss-minimizing output and incurs a loss that is less than total fixed cost
- By producing 7 sweaters a day {profit maximizing level of output}, firm incurs ATC of \$20.14 and AVC of \$17, with loss being \$22 a day which equals TFC

Firm's Supply Curve

FIGURE 12.5 A Firm's Supply Curve



(a) Marginal cost and average variable cost



- A perfectly competitive firm's supply curve shows how its profit-maximizing output varies as the market price varies, other things remaining the same
- The supply curve is derived from the firm's marginal cost curve and average variable cost curves
- If the price rises, the firm increases its output—it moves up along its marginal cost curve
- When the price is less than minimum AVC (less than \$17 a sweater), the firm maximizes profit by temporarily shutting down

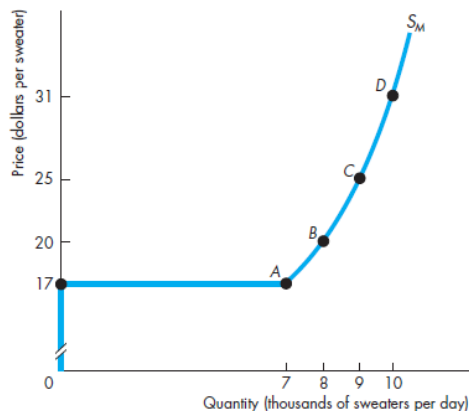
Output, Price and Profit in the Short Run

Market Supply in the Short Run

- The short-run market supply curve shows the quantity supplied by all the firms in the market at each price and the number of firms remain the same
- The market supply curve is derived from the individual supply curves and is a graph of the market supply schedules
- The quantity supplied by the market at a given price is the sum of the quantities supplied by all the firms in the market at that price
- Therefore, to construct the market supply curve, we sum the quantities supplied by all the firms at each price

Market Supply in the Short Run

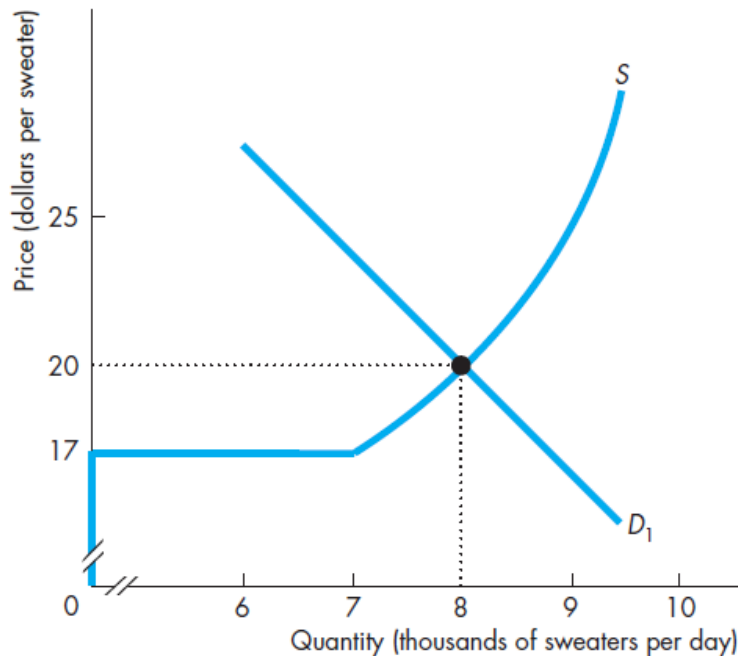
FIGURE 12.6 Short-Run Market Supply Curve



	Price (dollars per sweater)	Quantity supplied by Campus Sweaters (sweaters per day)	Quantity supplied by market (sweaters per day)
A	17	0 or 7	0 to 7,000
B	20	8	8,000
C	25	9	9,000
D	31	10	10,000

- For example, the market consists of 1,000 firms and at each price, the quantity supplied by the market is 1,000 times the quantity supplied by a single firm
- At price below \$17, every firm in the market shuts down and the quantity supplied by the market is zero
- At \$17 a sweater, each firm is indifferent between shutting down or operating (quantity supplied by each firm is either 0 or 7 sweaters)
- Above \$17 a sweater, each firm increases its quantity supplied and the quantity supplied by the market increases by 1,000 times that of one firm

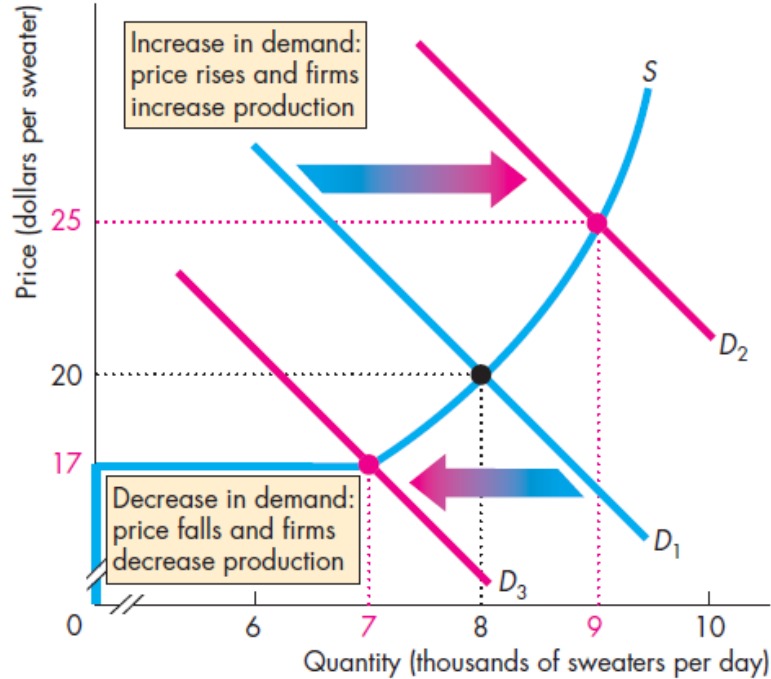
Short-Run Equilibrium



(a) Equilibrium

- Market demand and short-run market supply determine the market price and market output
- If the market demand curve is D_1 , the market price is \$20 a sweater
- Each firm takes this price as given and produces its profit-maximizing output, which is 8 sweaters a day
- Because the market has 1,000 identical firms, the market output is 8,000 sweaters a day

A Change in Demand



(b) Change in equilibrium

Profits and Losses in the Short Run

In the short run, the firm might break even (make zero economic profit), make an economic profit, or incur an economic loss – three possible outcomes

1. When the price equals minimum average total cost at the profit-maximizing level of output, the firm breaks even and makes zero economic profit
2. When the price exceeds average total cost at the profit-maximizing output, the firm makes an economic profit
3. When at the profit maximizing level of output the price is below minimum average total cost, the firm incurs an economic loss

The short run equilibrium corresponds to changes in the market demand

Expressing profits

If $TR = P \times Q$ and $TC = ATC \times Q$

Then, profits can be expressed in the following way

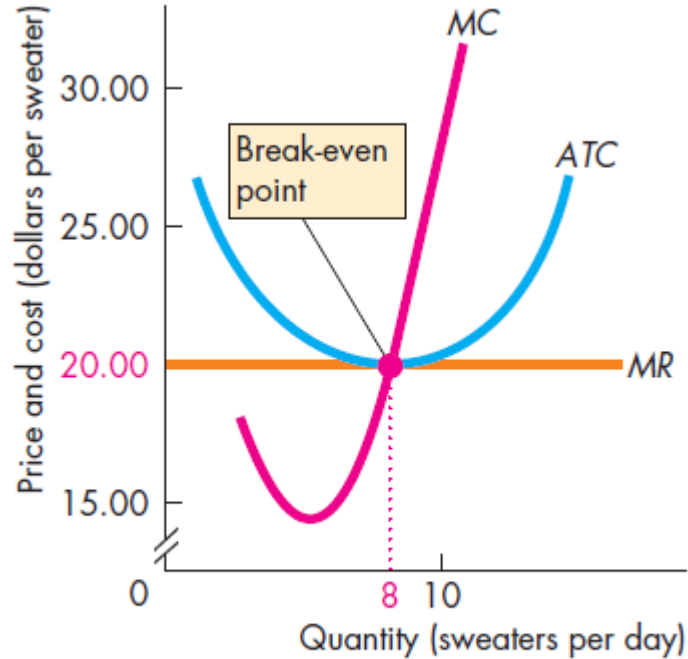
$$\text{Profit} = TR - TC$$

$$= (P \times Q) - (ATC \times Q)$$

$$= (P - ATC) \times Q$$

Therefore, profits > 0 when $P > ATC$

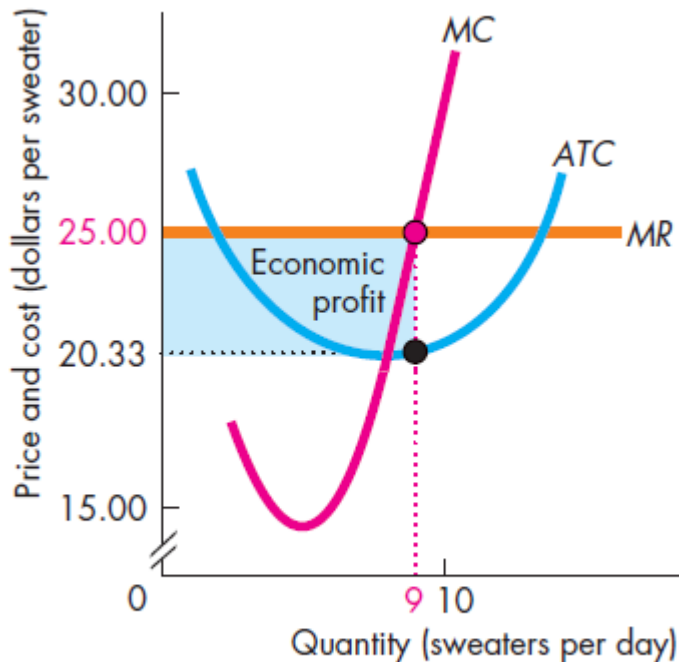
Short Run Outcomes - Break even



(a) Break even

- The equilibrium price of a sweater is \$20 and the firm produces 8 sweaters a day
- Average total cost is \$20 a sweater
- Price equals average total cost (ATC), so the firm breaks even (makes zero economic profit)

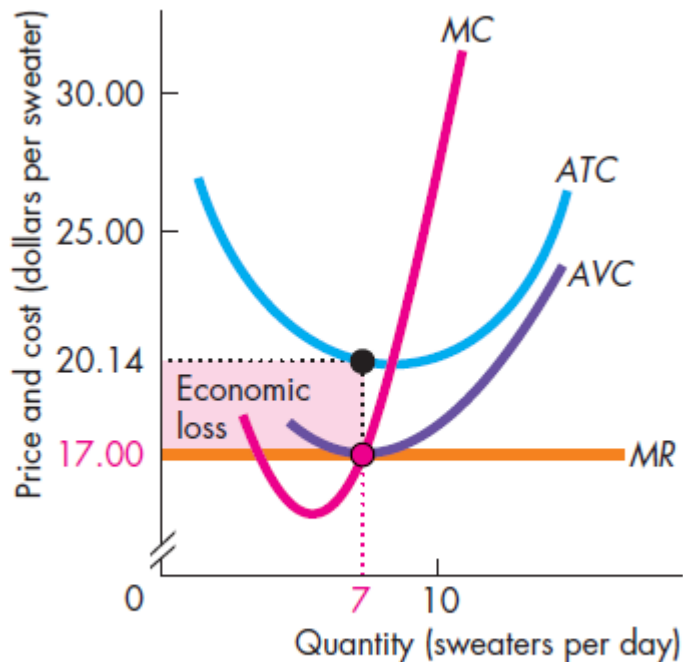
Short Run Outcomes - Economic Profit



(b) Economic profit

- The equilibrium price of a sweater is \$25 and the firm produces 9 sweaters a day (profit maximizing level of output)
- Here, price exceeds average total cost, so the firm makes an economic profit.
- Its economic profit is $\{(\$25.00 - \$20.33) \times 9\} = \$42$ a day

Short Run Outcomes - Economic Loss



(c) Economic loss

- The equilibrium price of a sweater is \$17
- Here, the price is less than average total cost, so the firm incurs an economic loss
- Price and marginal revenue are \$17 a sweater, and the profit-maximizing (in this case, loss minimizing) level of output is 7 sweaters a day
- Its economic loss is $\{(\$20.14 - \$17) \times 7\} = \$21.98$ a day

Output, Price and Profit in the Long Run

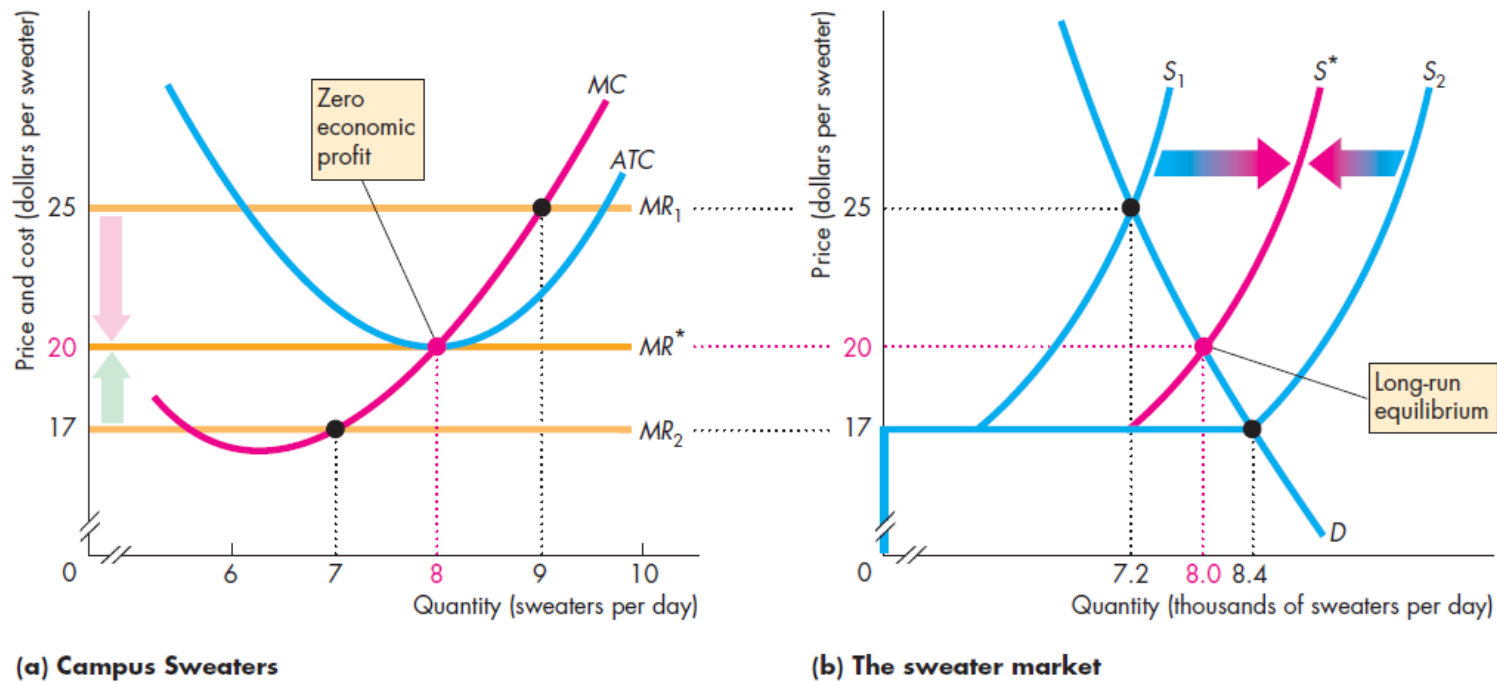
Entry and Exit

- New firms enter a market in which existing firms are making an economic profit
- As new firms enter a market, the market price falls and the economic profit of each firm decreases
- Firms exit a market in which they are incurring an economic loss
- As firms leave a market, the market price rises and the economic loss incurred by the remaining firms decreases
- Entry and exit stop when firms make zero economic profit

Note: Temporary economic profit and temporary economic loss do not trigger entry and exit. It's the prospect of persistent economic profit or loss that triggers entry and exit

Entry, Exit and Long Run Equilibrium

FIGURE 12.9 Entry, Exit, and Long-Run Equilibrium



Entry

- Suppose, the sweater market has 800 firms and each firm makes an economic profit
- This economic profit is a signal for new firms to enter the market
- As entry takes place, supply increases and the market supply curve shifts rightward toward S^* , with no change in demand
- As a result, the market price gradually falls from \$25 to \$20 a sweater and at this lower price, each firm makes zero economic profit and entry stops
- Because the price falls, each firm moves down its supply curve and produces less
- Because the number of firms increases, the market produces more
- Therefore, entry results in an increase in market output, but each firm's output decreases

Exit

- Suppose the sweater market has 1,200 firms and each firm incurs an economic loss
- This economic loss is a signal for firms to exit the market
- As exit takes place, supply decreases and the market supply curve shifts leftward toward S^* with no change in demand
- As a result, the market price gradually rises from \$17 to \$20 a sweater and at this higher price, losses are eliminated, each firm makes zero economic profit, and exit stops
- Because the price rises, each firm moves up its supply curve and produces more
- Because the number of firms decreases, the market produces less
- Exit results in a decrease in market output, but each firm's output increases

Long Run Equilibrium

- Economic profit induces entry, which in turn eliminates the profit and economic loss induces exit, which in turn eliminates the loss
- When economic profit and economic loss have been eliminated and entry and exit have stopped, a competitive market is in long-run equilibrium

Exercise

In a competitive market for bottled water, 200 firms operate selling identical products. The market price is Tk.15. Each firm has the following cost functions

$$TC = 40 + 4Q + Q^2$$

$$MC = 4 + 2Q$$

1. What is the profit maximizing level of output?
2. Calculate the firm's short-run profit.
3. Based on your answer to 2, comment whether the firm should shut down or continue to operate.
4. What happens in the long run if the losses were to continue?